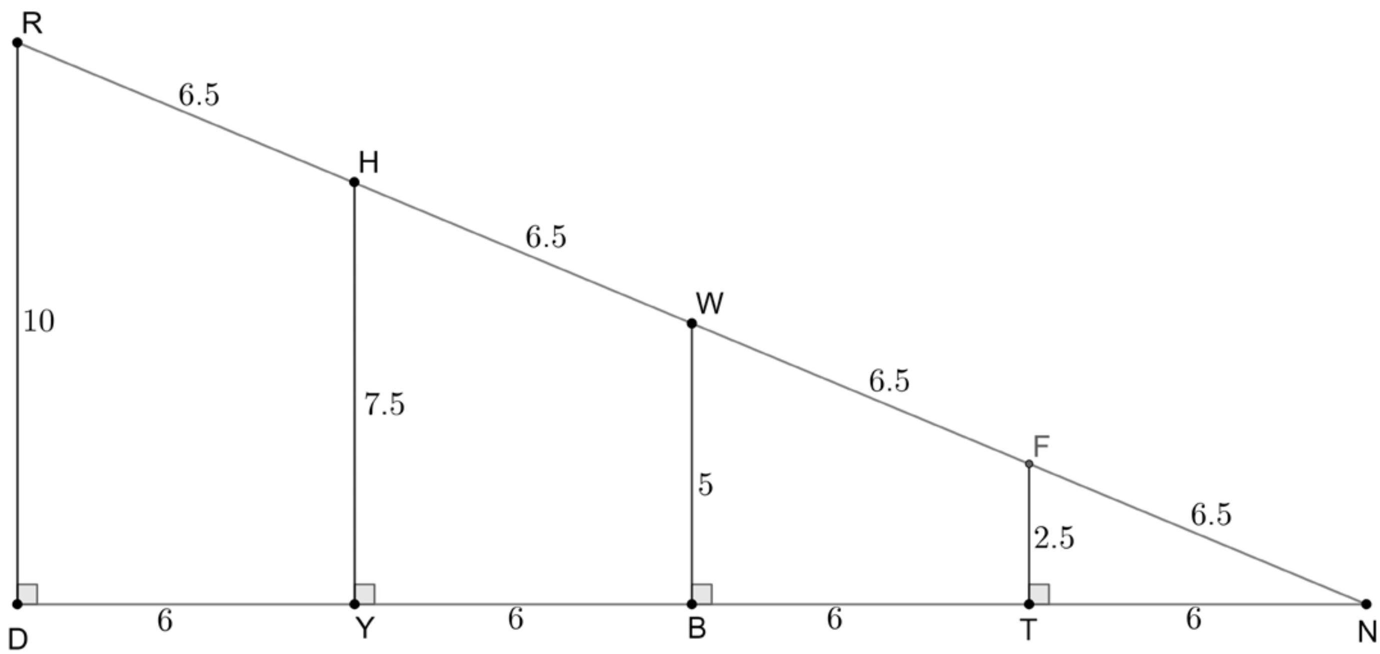
Geometry
Unit 7
Lesson 2 Homework

Name_Answer Key_________
Date____________________
Period___________________

The diagram shows measurements for $\triangle RND$, $\triangle HNY$, $\triangle WNB$, and $\triangle FNT$.



Complete the table using $\angle N$ as the reference angle.

	Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
1.	$\triangle FNT$	$\frac{2.5}{6.5}$	$\frac{6}{6.5}$	$\frac{2.5}{6}$
2.	$\triangle WNB$	$\frac{5}{13}$	$\frac{12}{13}$	$\frac{5}{12}$
3.	$\triangle HNY$	$\frac{7.5}{19.5}$	$\frac{18}{19.5}$	$\frac{7.5}{18}$
4.	$\triangle RND$	$\frac{10}{26}$	$\frac{24}{26}$	$\frac{10}{24}$

For $\triangle FNT$, $\triangle WNB$, $\triangle HNY$, and $\triangle RND$ find the decimal equivalent for each ratio.

	Ratio	$\triangle FNT$	$\triangle WNB$	$\triangle HNY$	$\triangle RND$
5.	$\frac{\textbf{opposite}}{\textbf{hypotenuse}}$	0.385	0.385	0.385	0.385
6.	$\frac{\textbf{adjacent}}{\textbf{hypotenuse}}$	0.923	0.923	0.923	0.923
7.	$\frac{\textbf{opposite}}{\textbf{adjacent}}$	0.417	0.417	0.417	0.417

8. Complete the statements.

The decimal equivalents of the $\frac{\text{opposite}}{\text{hypotenuse}}$ ratios for triangles FNT, WNB, HNY, and RND are

- o equal.
- o not equal.

The decimal equivalents of the $\frac{\text{adjacent}}{\text{hypotenuse}}$ ratios for triangles FNT, WNB, HNY, and RND are

- o equal.
- o not equal.

The decimal equivalents of the $\frac{\text{opposite}}{\text{adjacent}}$ ratios for triangles FNT, WNB, HNY, and RND are

- o equal.
- o not equal.

9. Complete the statement.

Since triangles FNT, WNB, HNY, and RND are

- o acute
- o obtuse
- o right

triangles that share

- o $\angle N$
- o side $\overline{DN}$
- o sides $\overline{DN}$ and $\overline{RN}$

the triangles are

- o congruent.
- o similar.

10. Complete the statement.

Because triangles FNT, WNB, HNY, and RND are

- o congruent
- o similar

their side lengths are

- o congruent.
- o proportional.